REVISION 2021 • SEMESTER 6 • TEXTILE TECHNOLOGY

Total Quality Management

A full-screen, syllabus-style digital handbook for Course Code 6062A. It is written as a practical textbook with module explanations, Malayalam support notes, formulas, checklists, worked cases, lab/report guidance, revision questions, model question paper and full answer key.

COURSE CODE
6062A

COURSE TITLE
Total Quality Management

DEPARTMENT
Textile Technology

TYPE
Theory

SEMESTER
6

CATEGORY
Open Elective

USE
Study + notes PDF

FORMAT
Big handbook

6062A

Screen-filling handbook

Designed for desktop, mobile, classroom display and automatic PDF notes generation.

How to use this handbook

Read the overview first, then study each module with the diagram, formulas, practical tasks and model answers. For exam preparation, revise the question bank and answer key.

Course map	**objectives + outcomes**
Module lessons	**4 detailed units**
Formula bank	**calculations**
Practice bank	**questions + tasks**
Model paper	**official style**
Answer key	**full points**

Course objectives and syllabus map

Objectives

- ✓ Understand TQM principles and customer-focused quality
- ✓ Use seven QC tools, 5S, Kaizen and PDCA
- ✓ Prepare quality improvement plan for textile processes

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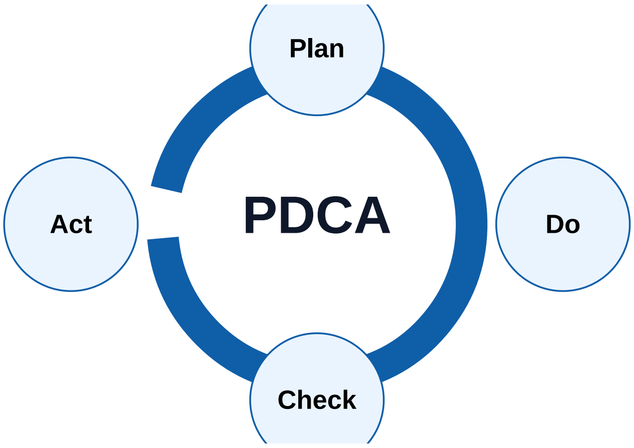
Visual process map



Module	Main outcome	Industrial focus	Expected evidence
M1	Quality and TQM concepts	Meaning of quality, quality control, quality assurance and TQM; Customer satisfaction, internal customer and external customer	Short notes, calculations, diagram, checklist, viva answers
M2	QC tools and problem solving	Seven QC tools: check sheet, histogram, Pareto, cause-effect, scatter, control chart and stratification; PDCA cycle and DMAIC idea	Short notes, calculations, diagram, checklist, viva answers
M3	Workplace quality systems	5S, Kaizen, standard operating procedures and visual management; ISO 9001 fundamentals, documentation and audit trail	Short notes, calculations, diagram, checklist, viva answers
M4	TQM implementation	Quality policy, measurable objectives and process indicators; Customer complaint analysis and corrective/preventive action	Short notes, calculations, diagram, checklist, viva answers

Module-wise textbook

Each module is written in clear engineering language. Do not mug up headings only; understand the process flow, defects, records and decision points.

Module 1

Quality and TQM concepts

Core explanation

- ✓ Meaning of quality, quality control, quality assurance and TQM
- ✓ Customer satisfaction, internal customer and external customer
- ✓ Quality gurus: Deming, Juran, Crosby and Ishikawa
- ✓ Cost of quality and prevention-based thinking

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
2. Draw block diagram/process flow wherever possible.
3. List parameters, machines/tools, defects and precautions.
4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 2

QC tools and problem solving

Core explanation

- ✓ Seven QC tools: check sheet, histogram, Pareto, cause-effect, scatter, control chart and stratification
- ✓ PDCA cycle and DMAIC idea
- ✓ Root cause analysis and 5 Why method
- ✓ Data-based decision-making in textile production

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

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- Missing units in calculations.

- Ignoring quality records and inspection points.
- Not connecting the topic to textile production or customer requirement.

Module 3

Workplace quality systems

Core explanation

- ✓ 5S, Kaizen, standard operating procedures and visual management
- ✓ ISO 9001 fundamentals, documentation and audit trail
- ✓ Training, teamwork, quality circles and suggestion schemes
- ✓ Supplier quality and incoming material control

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Module 4

TQM implementation

Core explanation

- ✓ Quality policy, measurable objectives and process indicators
- ✓ Customer complaint analysis and corrective/preventive action
- ✓ Benchmarking, reliability and continual improvement
- ✓ Applying TQM to spinning, weaving, processing and apparel units

Industry notes

In a real textile unit, this module connects with production planning, material control, quality assurance, cost control and customer requirement. A student should be able to explain what is checked, why it is checked, how it is recorded and what action is taken when the result is not acceptable.

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Exam writing method

1. Start with definition or purpose.
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4. End with application or quality importance.

Common mistakes

- Writing only theory without industrial example.
- Missing units in calculations.
- Ignoring quality records and inspection points.

- Not connecting the topic to textile production or customer requirement.

Formula bank and quick calculations

Formula 1

Defect % = Defective Units / Total Units x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 2

First Pass Yield = Good Units Without Rework / Total Units x 100

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 3

Quality Cost = Prevention + Appraisal + Internal Failure + External Failure

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Formula 4

Pareto Rule: focus on vital few causes

Use correct units, write given data first, substitute values and state the final conclusion clearly.

Practice tasks, viva and revision

Practical / assignment tasks

- ✓ Prepare Pareto chart for fabric defects
- ✓ Draw fishbone diagram for shade variation
- ✓ Prepare 5S audit checklist
- ✓ Write CAPA for a customer complaint

Important keywords

TQM **PDCA** **Kaizen** **5S** **Pareto** **Fishbone** **ISO 9001** **CAPA**

Short questions

Q1.1	Explain: Meaning of quality, quality control, quality assurance and TQM	Answer with definition, process, application and one example.
Q1.2	Explain: Customer satisfaction, internal customer and external customer	Answer with definition, process, application and one example.
Q1.3	Explain: Quality gurus: Deming, Juran, Crosby and Ishikawa	Answer with definition, process, application and one example.
Q2.1	Explain: Seven QC tools: check sheet, histogram, Pareto, cause-effect, scatter, control chart and stratification	Answer with definition, process, application and one example.
Q2.2	Explain: PDCA cycle and DMAIC idea	Answer with definition, process, application and one example.
Q2.3	Explain: Root cause analysis and 5 Why method	Answer with definition, process, application and one example.
Q3.1	Explain: 5S, Kaizen, standard operating procedures and visual management	Answer with definition, process, application and one example.
Q3.2	Explain: ISO 9001 fundamentals, documentation and audit trail	Answer with definition, process, application and one example.

Q3.3	Explain: Training, teamwork, quality circles and suggestion schemes	Answer with definition, process, application and one example.
Q4.1	Explain: Quality policy, measurable objectives and process indicators	Answer with definition, process, application and one example.
Q4.2	Explain: Customer complaint analysis and corrective/preventive action	Answer with definition, process, application and one example.
Q4.3	Explain: Benchmarking, reliability and continual improvement	Answer with definition, process, application and one example.

Case study

A textile unit receives customer complaint related to TQM. Prepare a corrective action report using observation, data collection, root cause, corrective action, preventive action and verification.

Expected answer structure: Problem statement → data/table → cause analysis → action plan → responsible person → completion date → verification evidence.

Model question paper

Use this as a sample paper. Questions are original and arranged in diploma-style sections.

PART A - Short Answer Questions (answer all)
1. Define Total Quality Management.
2. List four important keywords: TQM, PDCA, Kaizen, 5S.
3. State two industrial applications of this subject.
4. Mention two common defects/problems and one preventive action.
5. Write any one important formula or calculation used in this subject.

PART B - Paragraph Questions
6. Explain the workflow of Quality and TQM concepts.
7. Compare two important concepts from QC tools and problem solving.
8. Explain the practical importance of Workplace quality systems in a textile unit.
9. Write the steps for documentation/reporting in this subject.

PART C - Essay / Application Questions
10. Prepare a complete process plan for Total Quality Management in a textile industry situation.
11. Explain how quality, cost, safety and customer requirement are controlled in this subject.
12. Solve/interpret a simple industrial case using the formulas and checks given in this handbook.

Answer key and scoring points

Part A answer points

1. Give accurate definition and scope of Total Quality Management.
2. Use keywords: TQM, PDCA, Kaizen, 5S, Pareto, Fishbone, ISO 9001, CAPA.
3. Applications must be linked to textile industry, customer requirement and quality.
4. Defects/problems should include cause and prevention.
5. Formula answers must show unit and interpretation.

Part B/C - Module 1

Quality and TQM concepts

- Meaning of quality, quality control, quality assurance and TQM
- Customer satisfaction, internal customer and external customer
- Quality gurus: Deming, Juran, Crosby and Ishikawa
- Cost of quality and prevention-based thinking

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 2

QC tools and problem solving

- Seven QC tools: check sheet, histogram, Pareto, cause-effect, scatter, control chart and stratification
- PDCA cycle and DMAIC idea
- Root cause analysis and 5 Why method
- Data-based decision-making in textile production

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 3

Workplace quality systems

- 5S, Kaizen, standard operating procedures and visual management
- ISO 9001 fundamentals, documentation and audit trail
- Training, teamwork, quality circles and suggestion schemes
- Supplier quality and incoming material control

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Part B/C - Module 4

TQM implementation

- Quality policy, measurable objectives and process indicators
- Customer complaint analysis and corrective/preventive action
- Benchmarking, reliability and continual improvement
- Applying TQM to spinning, weaving, processing and apparel units

Scoring: concept 2 marks, diagram/process 2 marks, application 2 marks, conclusion 1 mark.

Final quick revision

Before exam

- Revise all formulas and keywords.
- Practise one diagram/process flow from every module.
- Prepare two industrial examples for every long answer.

Before lab/viva

- Know instrument/machine purpose.
- Know safety precautions and standard record format.
- Explain result, defect and correction logically.